11–5 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11–5.

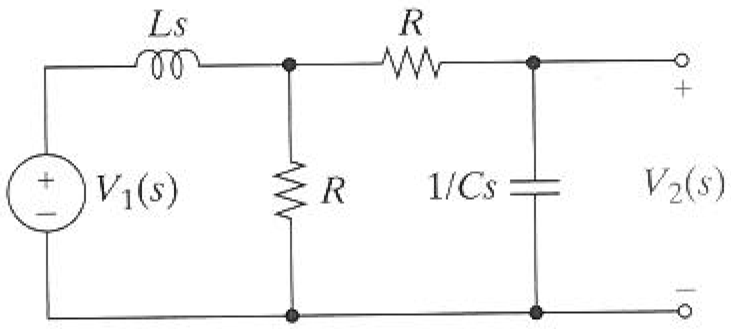


FIGURE P11–5

11–8 Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11–8. Select values of R and L so that $T_I(s)$ has a pole at $s = -500$ rad/s.

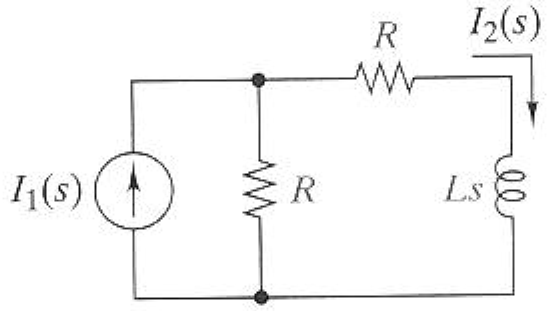


FIGURE P11–8

11–9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1($ of the cascade connection in Figure P11–9. Locate th poles and zeros of the transfer function.

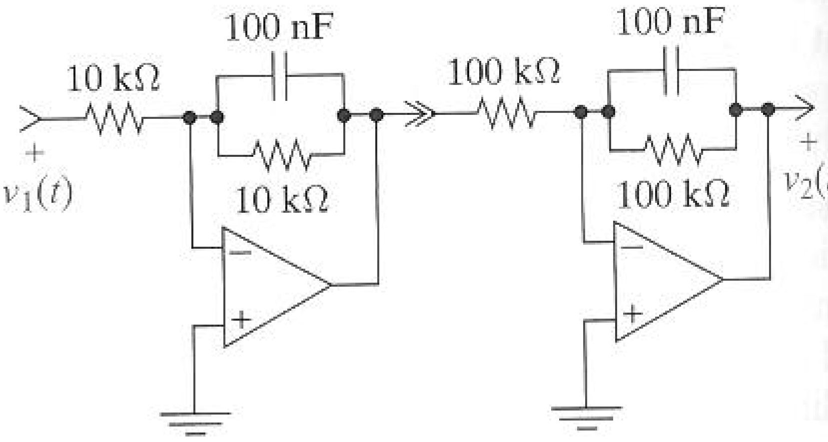


FIGURE P11–9

11–30 The circuit in Figure P11–30 is in the steady state with $i_1(t) = 10 \cos 5000t$ mA. Find the steady-state voltage $v_{2SS}(t)$. Repeat with $i_1(t) = 5 \cos 2500t$ mA.

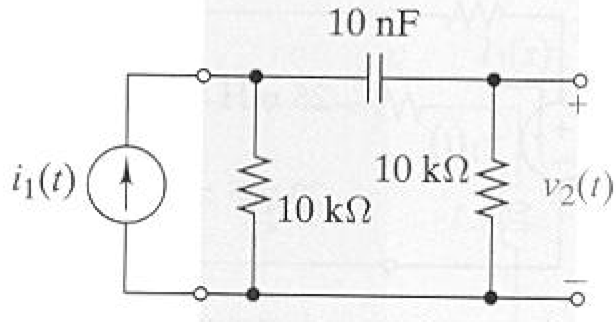


FIGURE P11–30

Design a circuit that realizes a given $T(s)$ and meets other stated constraints.
Evaluate alternative designs using stated criteria.

-51 **D** Design a circuit to realize the transfer function below using only resistors, capacitors, and OP AMPs. Scale the circuit so that all resistors are greater than 10 kΩ and all capacitors are less than 1 μF.

$$T_V(s) = \pm \frac{2 \times 10^4}{(s + 250)(s + 2500)}$$

-52 **D** Design a circuit to realize the transfer function below using only resistors, capacitors, and not more than one OP AMP. Scale the circuit so that all capacitors are exactly 10 nF.

$$T_V(s) = \pm \frac{100(s + 1000)}{(s + 250)(s + 2000)}$$

12–3 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12–3.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
(b) Sketch the straight-line approximation of the gain response.
(c) What element values would you change to increase the passband gain to 10?

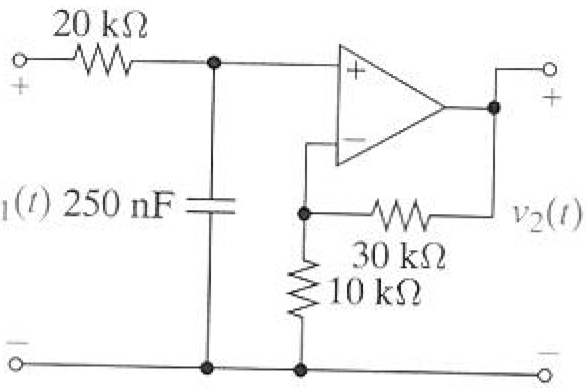


FIGURE P12–3

12–9 The transfer function of a first-order circuit is

$$T(s) = \frac{2}{10^{-2} + 20/s}$$

(a) Identify the type of gain response. Find the cutoff frequency and the passband gain.
(b) Sketch the straight-line gain response. Use the straight-line gain to estimate the gain at $\omega = 0.5\omega_C$, ω_C, and $2\omega_C$.

12–16 The circuit in Figure P12–16 produces a bandstop response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the cutoff frequencies are 10 rad/s and 500 rad/s. What is the passband gain for your choices? Use practical element values with $R \geq 10\ \text{k}\Omega$, $L \leq 100$ mH, and $C \leq 1\ \mu$F.

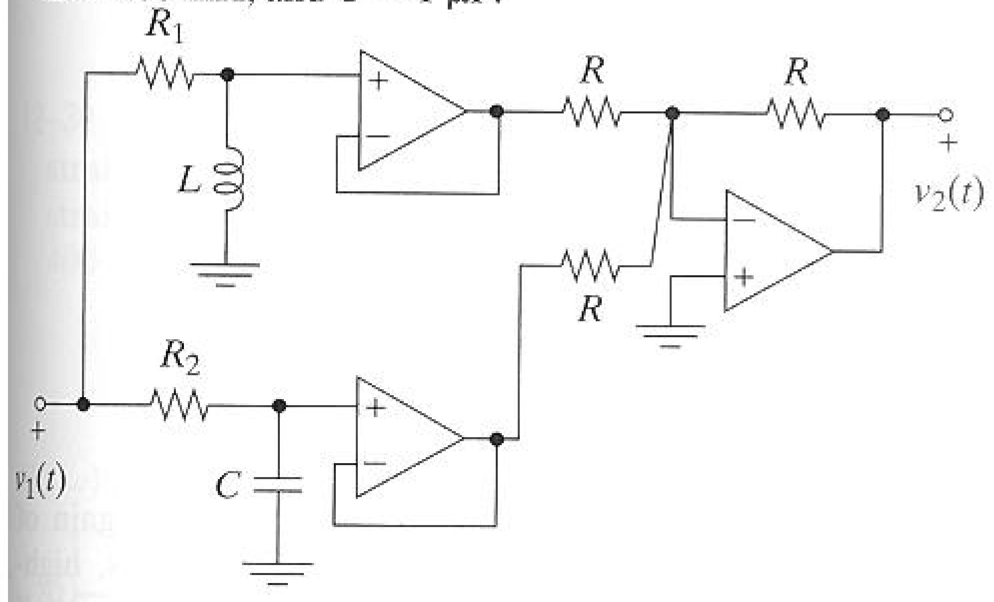


FIGURE P12–16

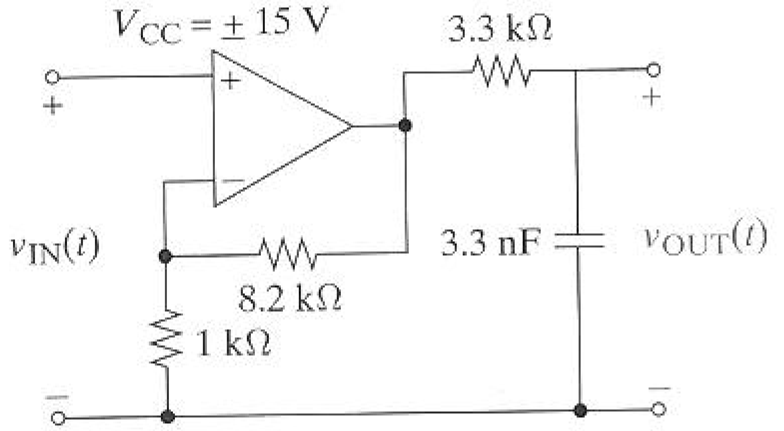
12–55 **E** Design Evaluation

Your company issued a request for proposals listing the following design requirements and evaluation criteria.

Design requirements: Design a low-pass filter with a pass-band gain of $9 \pm 10\%$ and a cutoff frequency of $90 \pm 10\%$ krad/s. The filter input is driven by a sensor with a 1-kΩ source resistance and an open-circuit voltage range of ±1.6 V.

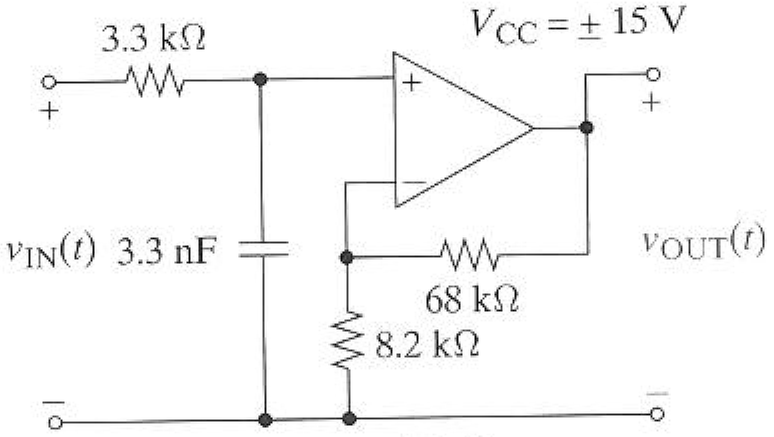
Evaluation criteria: Evaluate filter performance, parts count, use of standard parts, and cost. The two vendors have responded with the designs shown in Figure P12–55. As a junior engineer, you are asked to evaluate the designs and recommend a vendor. Which vendor would you recommend and why?

First-Order Filter Company



Unit Cost: $29.00

Simply Filters, Ltd



Unit Cost: $20.99

FIGURE P12–55